

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	(a)		<u>Continuous spectrum</u> due to radiation of all wavelengths emitted from surface of star [1] [superimposed] line <u>absorption</u> spectrum (due to passage of radiation) <u>through atmosphere</u> (of star) [1]	2			2		
	(b)	(i)	Use of $I = \frac{P}{4\pi R^2}$ [1] Substitution: $\frac{I_S}{I_V} = \frac{9.7 \times 10^{27} \times [4\pi] (2.4 \times 10^{17})^2}{1.5 \times 10^{28} \times [4\pi] (8.1 \times 10^{16})^2}$ [1] Alternative for 2nd mark: Calculation of either intensity: $I_{\text{Sirius}} = 1.18 \times 10^{-7} \text{ [W m}^{-2}\text{]}$ or $I_{\text{Vega}} = 2.07 \times 10^{-8} \text{ [W m}^{-2}\text{]}$ Ratio = 5.7 [1]	1	1 1		3	2	
		(ii)	Shape of curve below that of Sirius at all points [1] Peak at $(1, \lambda_{\text{max}})$ ecf [1]		2		2		
		(iii)	Substitution into $P=4\pi R^2 \sigma T^4$ i.e. $9.7 \times 10^{27} = 1.8 \times 10^{19} \times 5.67 \times 10^{-8} \times T^4$ [1] $T = 9.9 \times 10^3 \text{ K (9874 K)}$ [1] Substitution into $\lambda_{\text{max}} = \frac{W}{T} = \frac{2.90 \times 10^{-3} \text{ [m K]}}{9.9 \times 10^3 \text{ [K]}}$ ecf [1] $\lambda_{\text{max}} = 2.9 \times 10^{-7} \text{ [m]}$ [1]	1 1	1 1		4	4	

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	(c)		Reference to multiwavelength astronomy / using different parts of the em spectrum [1] Early photographs used visible light so limited / extra wavelengths provide extra information [1] Extra detail provided e.g. link between wavelength and temperature such as quasars at gamma / X-rays [1] This detail could be evidence for the 2 nd mark.			3	3		
			Question 7 total	5	6	3	14	6	0